

Brady Bangasser

brady@bangasser.dev | (651) 769-5301 | www.bangasser.dev |  BradyBangasser |  bbangasser | Saint Paul, MN

Systems engineer running production HPC clusters and researching high-performance and distributed computing, from scheduler internals to GPU-offload optimization at supercomputer scale.

EXPERIENCE

Iowa State University

Aug 2025 – Present

Research Assistant – Compiler Optimization

- Built OMPRTF (OpenMP RunTime Fabric) for Iowa State's SWAPP lab, a custom OpenMP runtime and dynamic-analysis pipeline that applies aggressive, conditional optimizations to GPU-offload programs by lowering them to LLVM IR.
- Wrote an LLVM optimization pass that breaks opaque OpenMP runtime calls into discrete IR instructions, pruning redundant host-GPU data movement without destabilizing kernels.

Iowa State University

Apr 2025 – Present

System Administrator

- Operate and maintain a production HPC cluster with SLURM, Kubernetes, Active Directory, Ansible, Prometheus, and Apptainer, keeping it secure and available for a large user base.
- Built diagnostic and monitoring workflows that isolate hardware degradation, network bottlenecks, and software-stack faults before they take nodes offline.

Iowa State University

Oct 2024 – May 2025

Research Assistant – Machine Learning

- Designed and trained object-classification models with PyTorch and OpenCV, tuning data preprocessing and tensor operations to cut training latency without losing accuracy.
- Tested perception models on self-driving vehicles to catch false and failed classifications.

Bethel University

Aug 2023 – Aug 2024

Research Assistant – High Performance Computing

- Built and configured two bare-metal HPC clusters from scratch: networking, custom DNS and routing, user management, and SLURM job scheduling with VMware and Debian.
- Implemented a wildfire simulation in OpenMP, CUDA, C, and Fortran to benchmark cluster performance across heterogeneous compute.
- Wrote a lightweight daemon and control program to manage per-node status lighting across the clusters.

EDUCATION

Iowa State University

Ames, IA

M.S. Computer Science · GPA: 4.00

Aug 2025 – May 2027

- Concurrent B.S./M.S. program
- Coursework: Advanced Topics in High Performance Computing, Distributed Systems and Middleware, Data Security in Machine Learning, Network Design and Security

Iowa State University

Ames, IA

B.S. Computer Science · GPA: 3.79 · Dean's List

Aug 2024 – Dec 2026

- Coursework: Compiler Design and Development, AI and Machine Learning in High Performance Computing, Network Architecture and Security, Theory of Computing, Software Design Practices

TECHNICAL SKILLS

Languages: C, C++, Python, Go, Rust, Bash, CUDA, Fortran

Infrastructure & Platform: Kubernetes, SLURM, Apptainer, Linux/Unix, Git

Compilers & HPC: LLVM, OpenMP, OpenMPI, PyTorch